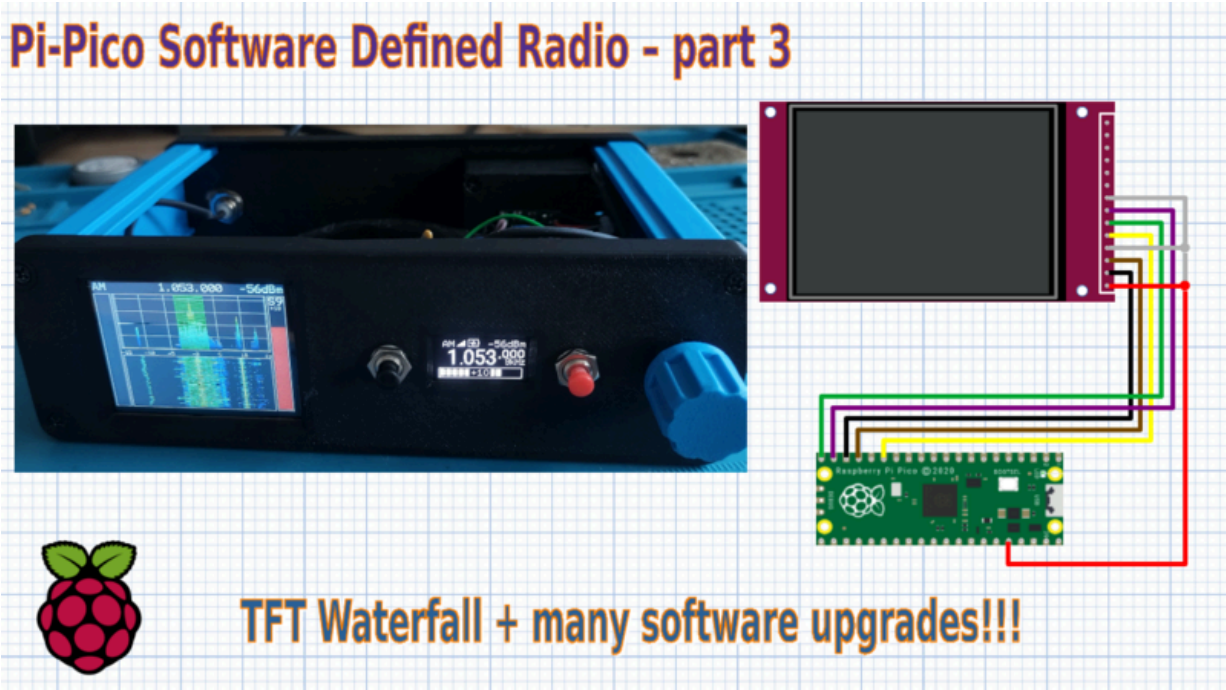
# Breadboard Radio - Part 3




## FFT-based filtering

One significant change in this release of firmware is the introduction of frequency-domain filtering using an FFT.

In the time domain, filtering is achieved by convolving a signal with a filter kernel containing the impulse response of a filter. To build better filters with narrow pass-bands and sharp transitions, we need to use a longer filter kernel. Convolution in the time domain is computationally expensive, for each sample, we need to multiply the previous N samples by a filter kernel N samples long, thus the number of multiplications required is proportional to the kernel length.

In the frequency domain, filtering is a much simpler operation that only requires each frequency sample to be multiplied by a single constant representing the magnitude of the frequency response. Of course, we first need to convert the signal from a time domain to a frequency domain and back again. We use the Fast Fourier Transform (FFT) and its close relation, the Inverse Fast Fourier Transform (IFFT) to transform between the time and frequency representations.

The FFT algorithm is so efficient, that even when the additional effort of transforming to a from the frequency domain is considered, it is still faster than time domain convolution for large filter kernels.

The *windowed overlap-add method* involves dividing the input signal into smaller, overlapping segments, applying filtering in the frequency domain to each segment, and then recombining (adding) these segments to reconstruct the filtered signal. This approach allows for efficient filtering while avoiding issues that arise with direct filtering on long signals.

### Key Steps Involved

1. Divide the Signal into Overlapping Segments. The input signal is split into shorter, overlapping segments using a windowing function.

2. Apply FFT to Each Segment. For each segment, the windowed signal is transformed into the frequency domain using the Fast Fourier Transform (FFT). The windowing function minimizes edge effects by smoothly tapering the signal at the boundaries of each segment. The window shape and overlap are so that the overlapped signals add to a constant value.

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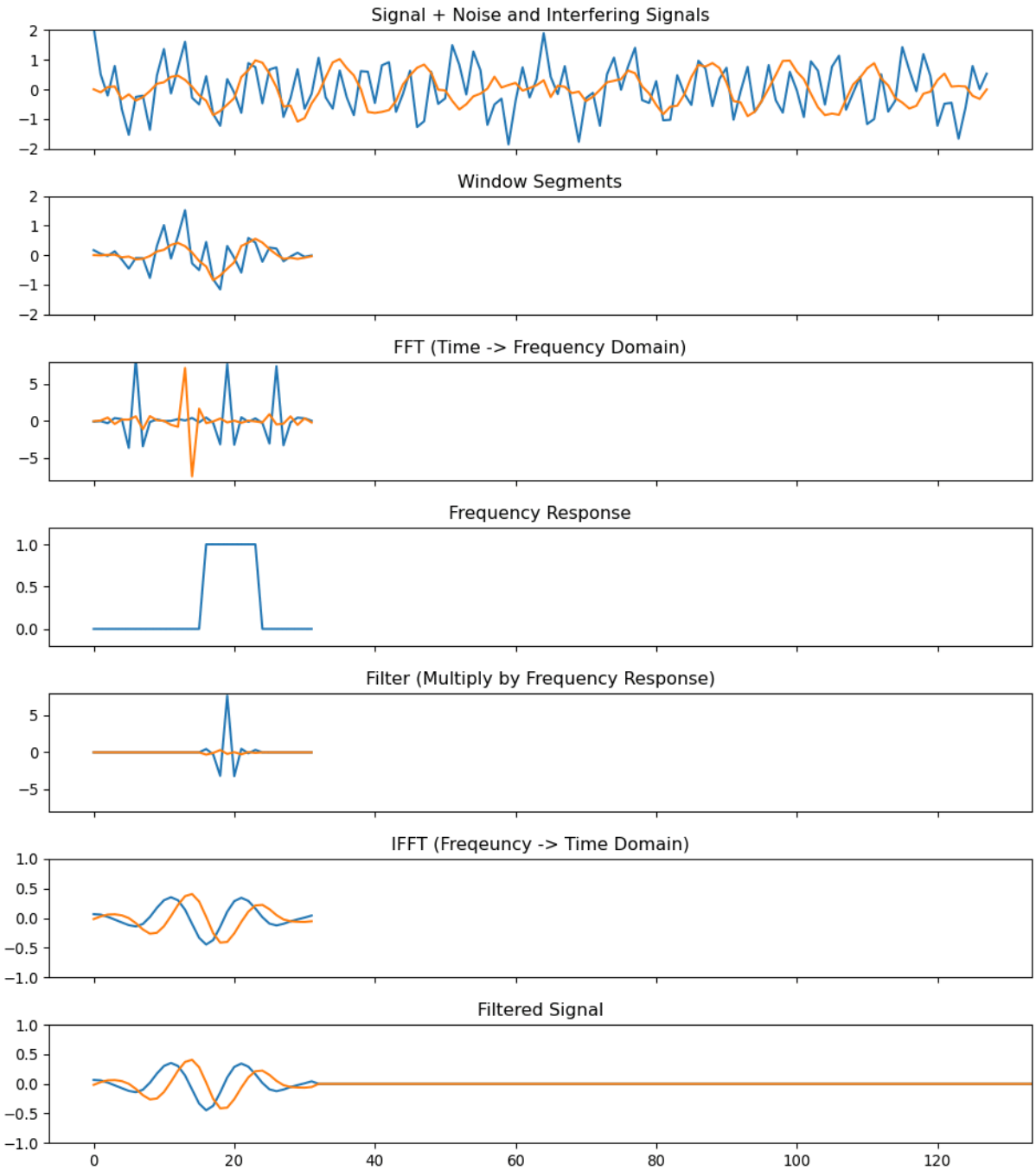
3. Multiply with the Frequency Response of the Filter. In the frequency domain, each segment is element-wise multiplied by the frequency response of the desired filter. This multiplication effectively filters the frequency content of the segment.

4. Inverse FFT (IFFT) to Convert Back to the Time Domain. Each filtered segment is converted back to the time domain using the Inverse FFT (IFFT). The result is a filtered version of the segment.

5. Reconstruct the Signal using Overlap-Add. The final step is to overlap and add the filtered segments to reconstruct the complete output signal. The overlapping segments are summed together at their overlapping portions. This ensures a smooth transition between segments without introducing discontinuities.




In this design, the first decimation stage is still performed using a CIC filter, since this provides a very efficient means to achieve a high decimation rate. In this design, a fourth-order CIC filter decimates by a factor of 16 reducing the sampling rate from 480kS/s to 30kS/s.

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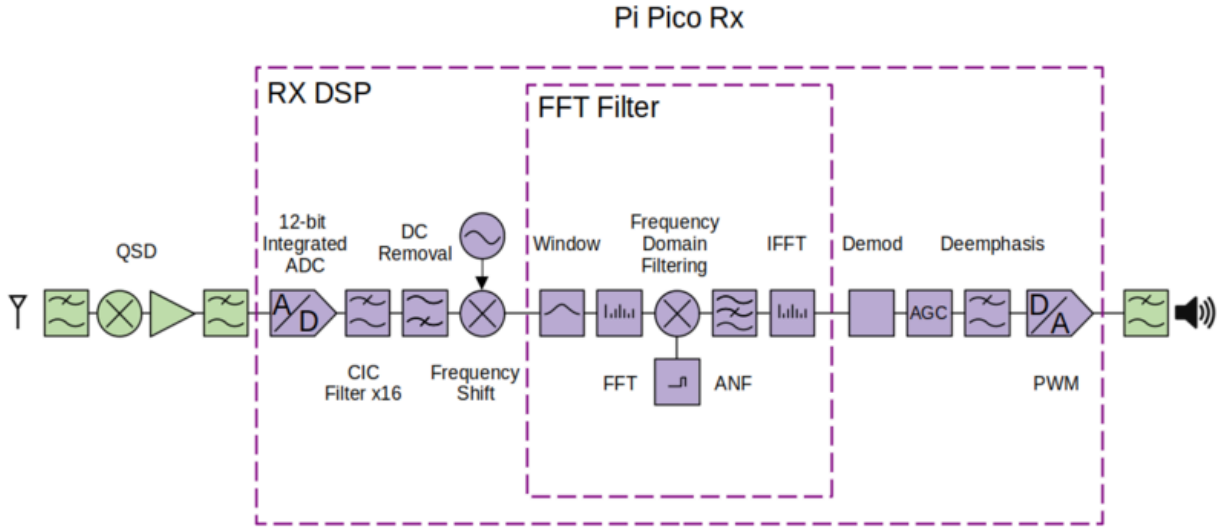


To maximise the performance of the receiver, while minimising the CPU load it is necessary to consider the bits growth through the FFT. A fixed point implementation is chosen using a 16-bit signed data word. A 16-bit word size is efficient because it allows a fixed point multiplication to be performed using a single 32-bit multiply which the pi-pico can perform in a single clock cycle.

The ADC outputs 12-bit data, but the ADC has an effective number of bits of 9.5 (for an RP2350 which fixes the ADC bug). This means that the lowest 2.5 bits contain noise rather than useful information. The CIC filter reduces the bandwidth by a factor of 16, which increases the effective number of bits by 2.

The FFT filter itself also reduces the bandwidth, taking 2.5kHz as a typical bandwidth for an SSB signal, the bandwidth is reduced by a factor of 3.6 giving an extra 1.9 useful bits. So, we might have about 13 useful bits once the filtering has been applied. To make use of the noise reduction during the FFT filter, the signal is first scaled by a factor of 2 giving 13 rather than 12 bits of data.

During the forward FFT, a growth of 256 can occur. With no scaling at all, a sin wave with an amplitude of 16 in the time domain would result in a peak with a magnitude of 4096 in the frequency domain. Various scaling strategies can be employed in an FFT to allow for this growth.

Scaling by a factor of 2 after each stage is safe and prevents any possibility of overflow. The downside of this approach is that we truncate the lowest 8 bits of the output data, 4 of these bits contain useful data and 4 contain noise.

To preserve as many useful bits as possible, a smaller scaling factor is used in this design. Scaling by a factor of 2 after every *second* stage means that we truncate by 4 bits overall, losing only the bits that contain noise. This means that there is a possibility of overflow under worst-case conditions if the input data exceeds 12 bits.

However, the worst-case growth can only occur for a single tone of large amplitude, and with our 13-bit input, the conditions needed to cause overflow are highly unlikely to occur in normal use.

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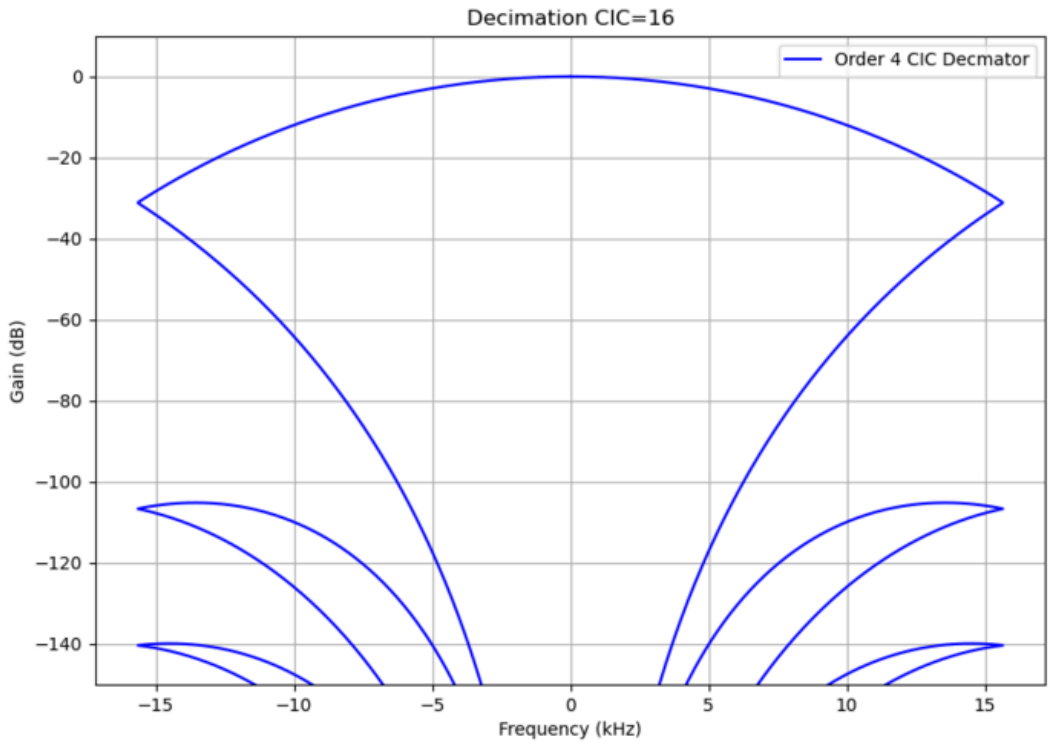
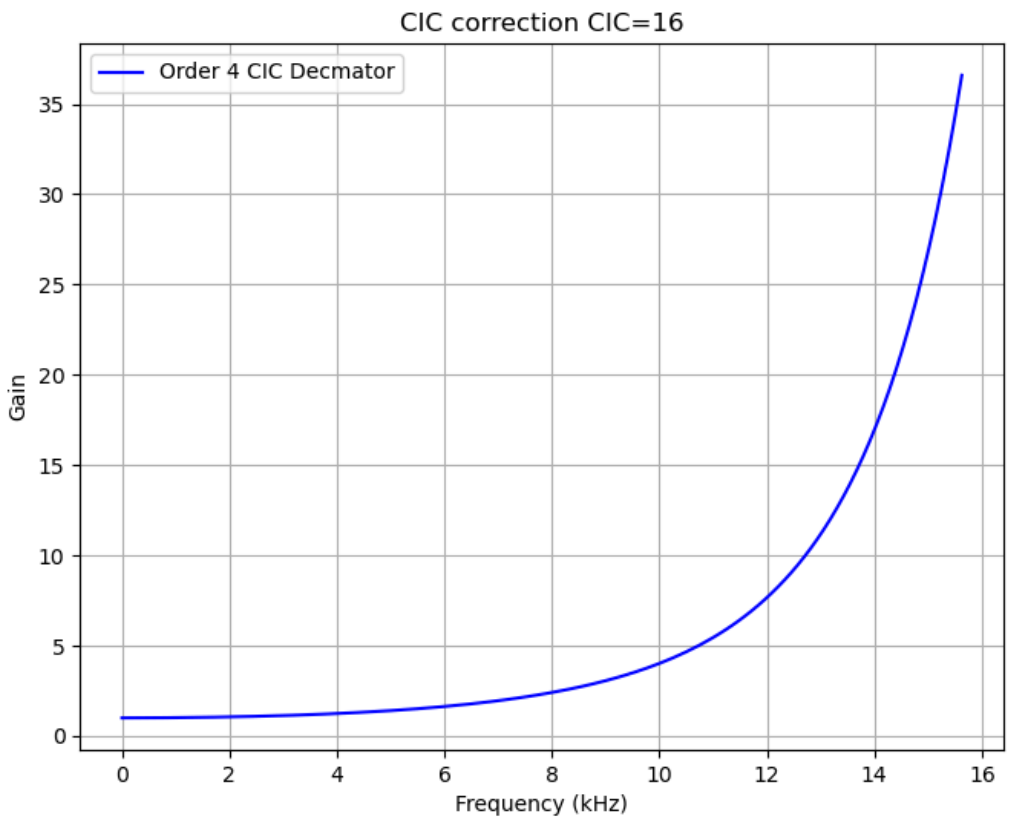



The frequency response of the CIC filter isn't particularly flat in the passband, fortunately, we can correct this in the FFT filter, by applying a gain to each bin which is the reciprocal of the CIC filter response.




An FFT with a length of 256 points is employed, this allows a sharp frequency response equivalent to a 256-tap FIR filter, and each frequency bin is 117Hz wide. The filter itself is very simple, frequency bi the pass band are retained (and adjusted to compensate for the CIC filter), while frequency bins in t

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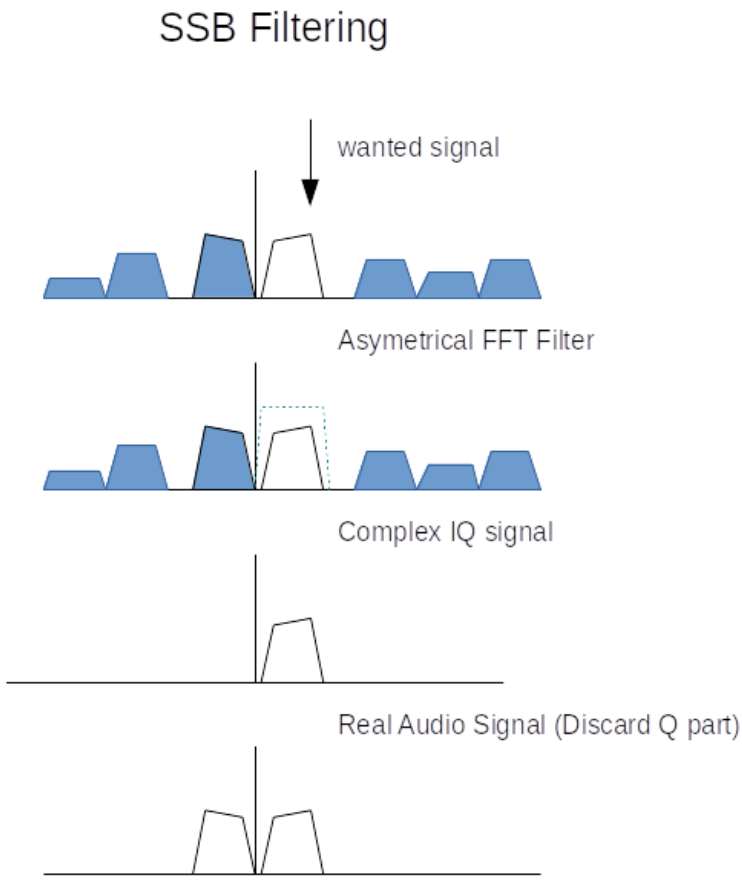
band are simply set to zero. It is now very simple to vary the bandwidth of the filter in real-time, simply by choosing how many bins to retain.

Another advantage of this approach is that it becomes very easy to implement asymmetric filters, which makes SSB filters very simple to implement. We simply remove the negative frequency bins for upper sideband signals and the positive frequency bins for lower sideband. Since we have efficiently removed the unwanted sideband there is no need for a separate Hilbert transform, demodulation of SSB signals then becomes a trivial matter of discarding the imaginary component of the signal.




Since we don't need the full 30kHz of bandwidth for the frequency modes we are working in, we can safely reduce the sample rate by another factor of 2 to reduce the computational load in the demodulator. This is simply achieved using a smaller 128-point IFFT returning the central part of the spectrum to the time domain.

After implementing the frequency domain filtering, the CPU loading hasn't increased significantly although the performance of the filters has been significantly improved.

# Automatic Notch Filter

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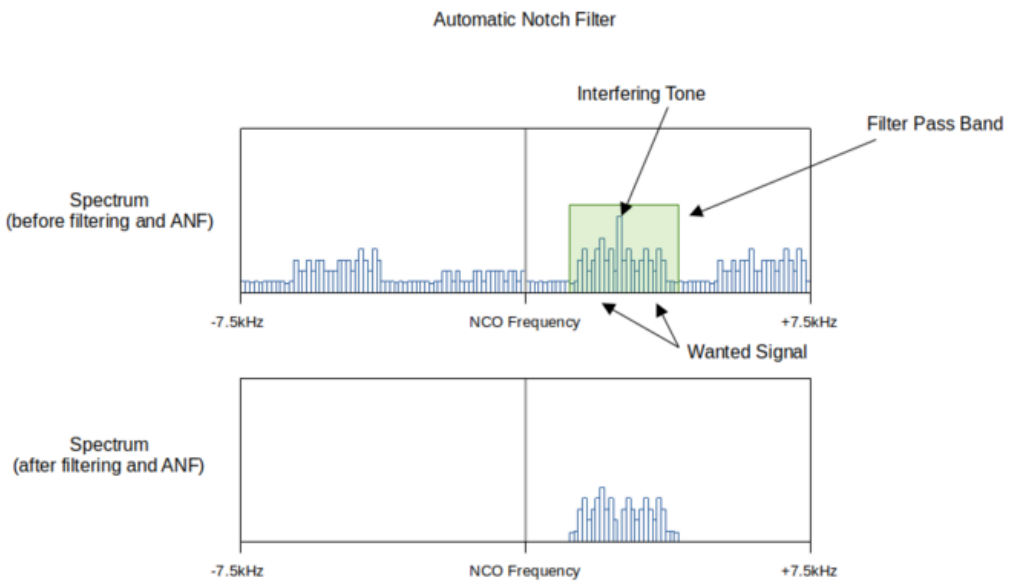


Since we are now filtering in the frequency domain, it is now possible to implement some more sophisticated DSP processing in the frequency domain without adding significant load to the CPU. An automatic notch filter can be used to remove unwanted CW interference.

The algorithm identifies CW interference by searching for the frequency bin with the highest magnitude. If the same bin is identified consistently over several frames, the frequency bin is attenuated removing the interfering signal.

The algorithm is very effective at removing an interfering tone from a speech signal.

**Interfering tone without ANF**

0:00 / 0:00

**Interfering tone with ANF**

0:00 / 0:00

# Synchronous Amplitude Modulation

Thanks to Mariusz Ryndzionek for contributing the synchronous AM demodulator!

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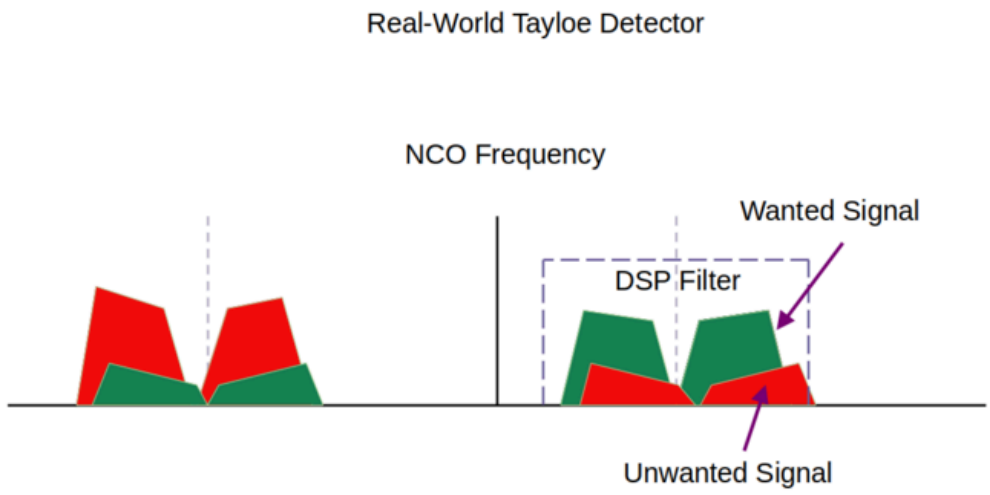


The existing AM demodulation is a very simple implementation that calculates the magnitude of the IQ baseband signal using a simple magnitude approximation. A more sophisticated method of AM demodulation is synchronous demodulation. Synchronous AM demodulation uses a Phase Locked Loop (PLL) to remove any frequency and phase offset.

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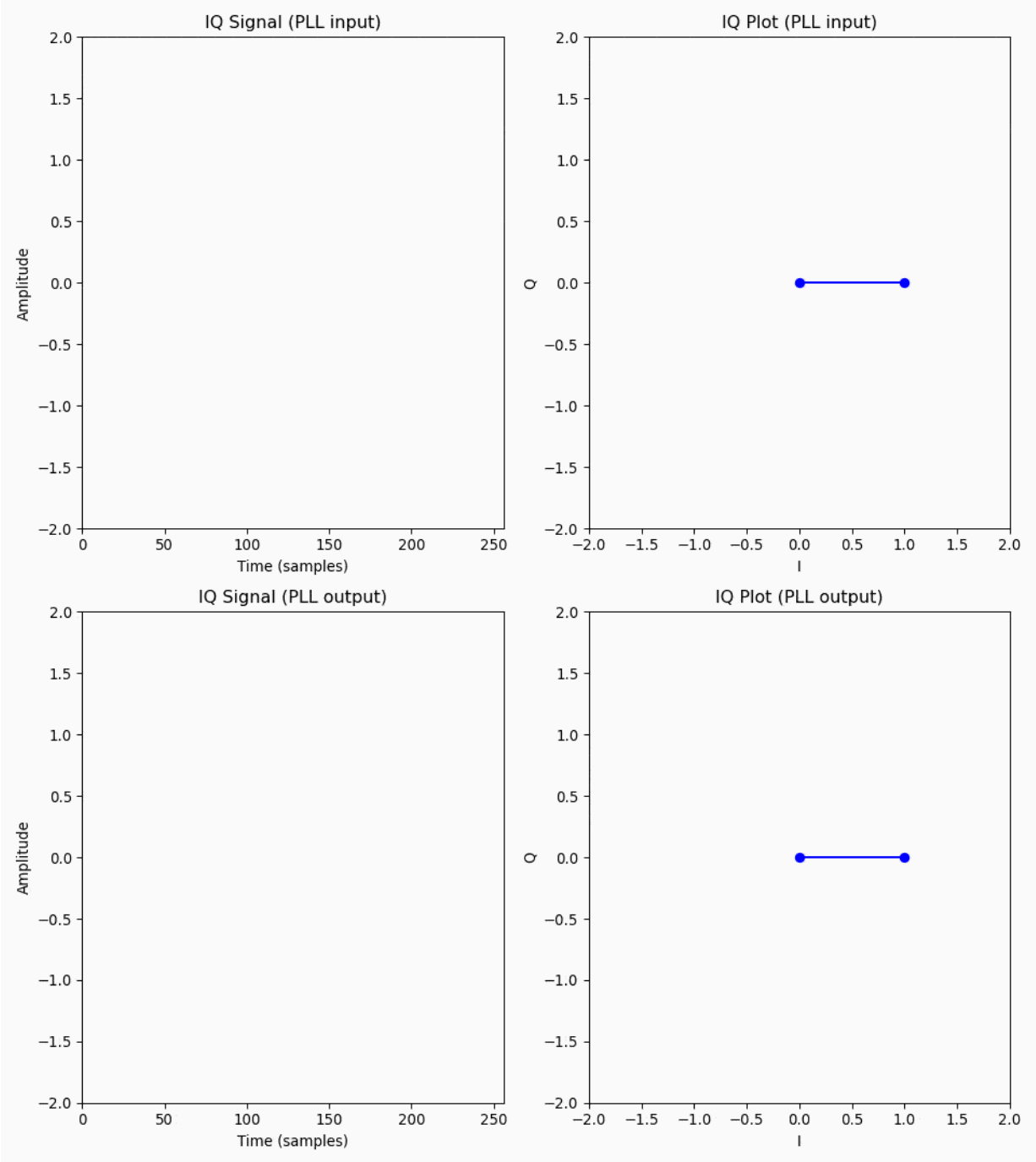


The frequency offset is generally caused by the inevitable small difference between the receiver's tuned frequency and the signal's carrier. After demodulation, the baseband signal has a steady zero frequency and phase offset. Since the phase is zero, the I component now contains all the information, and the C component is zero. Demodulation then becomes a simple matter of removing the DC offset in the I component.

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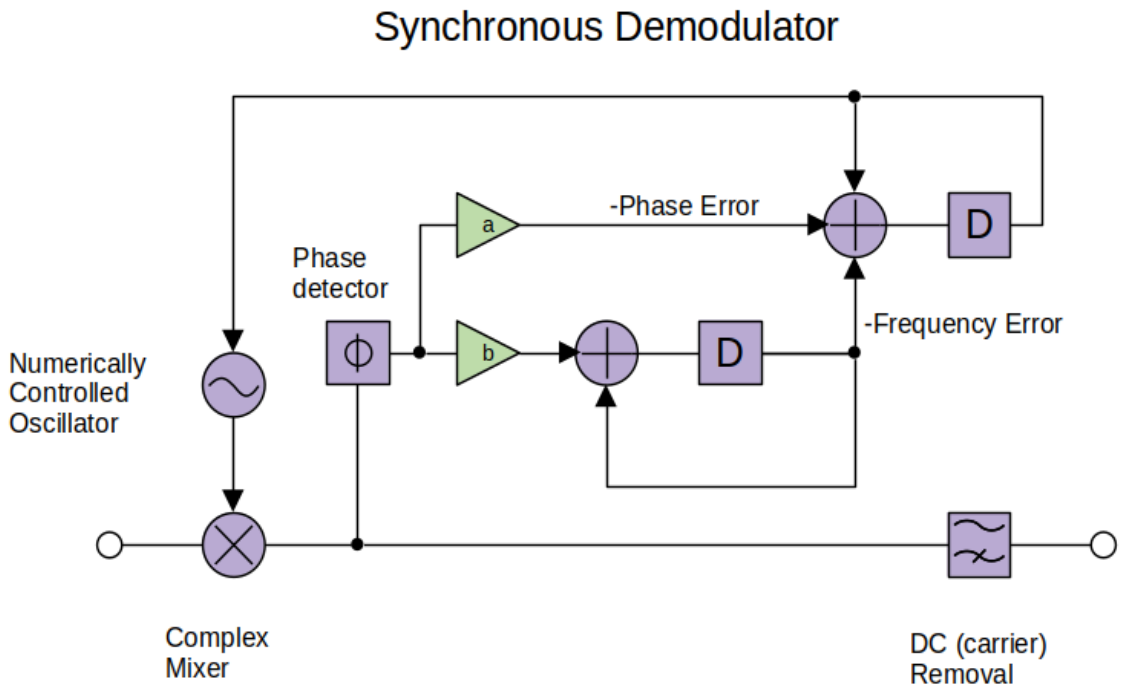


The Synchronous AM gives improved reception compared to the original implementation. I think that is in part because it avoids any distortion introduced by the magnitude approximation.

**AM Demodulator**

0:00 / 0:00

**Synchronous AM Demodulator**

0:00 / 0:00

# IQ Imblance Correction

Thanks to Mariusz Ryndzionek for contributing IQ imbalance correction!

One of the nice things about the QSD is that it is a complex mixer with the capability to reject images, in other words, it can distinguish positive frequencies (above the NCO) and negative frequencies (below the NCO).

Due to the limits of component tolerances in the real world, the I and Q inputs from the QSD have a small difference in magnitude and the phase difference isn't always 90 degrees. This manifests itself in a degradation in the ability to cancel positive and negative frequencies.

For example, when the wanted station is 6kHz above the QSD frequency, we might hear a signal 6kHz below the QSD frequency at a reduced level. This interfering signal is called an "image" or "mirror" signal.

The amount the mirror signal is attenuated is a figure of merit called image rejection. An easy way to measure the image rejection is to temporarily enable the "swap IQ" setting. This has the effect of swapping positive and negative frequencies. When tuning to a strong station, the signal may still be heard at a reduced level after enabling "swap IQ". The reduction in signal level when enabling the setting tells us the image rejection. I performed some baseline measurements of the image rejection in the receiver and the results are quite good.

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| Frequency | Image Rejection |
|---|---|
| 1.8 MHz | 42dB |
| 3.5 MHz | 42dB |
| 7 MHz | 34dB |
| 14 MHz | 37dB |
| 28 MHz | 38dB |

However, it is still possible to hear an image of a very strong signal, the image rejection can be improved using DSP techniques in software. The technique used is based on this paper by Mosley and Slump.

It works quite simply by estimating the difference in phase and magnitude between I and Q. The quantities theta1, theta2 and theta3 are accumulated over a block of data. This method of estimation is fast because the theta quantities are easy to calculate without using any of the trig functions that would normally be used to calculate the phase and magnitude of a signal.

```
theta1 += ((i < 0) ? -q : q);
theta2 += ((i < 0) ? -i : i);
theta3 += ((q < 0) ? -q : q);
```

Theta 1 represents the phase difference, it should give a value of zero when the I and Q signals are separated in phase by 90 degrees. Greater or smaller phase differences result in a positive or negative value. Theta 2 and theta 3 effectively rectify the I and Q signals giving a value proportional to their respective amplitudes. Ideally, we want the magnitudes to be the same. You can see the effect of phase and amplitude differences in the following animation.

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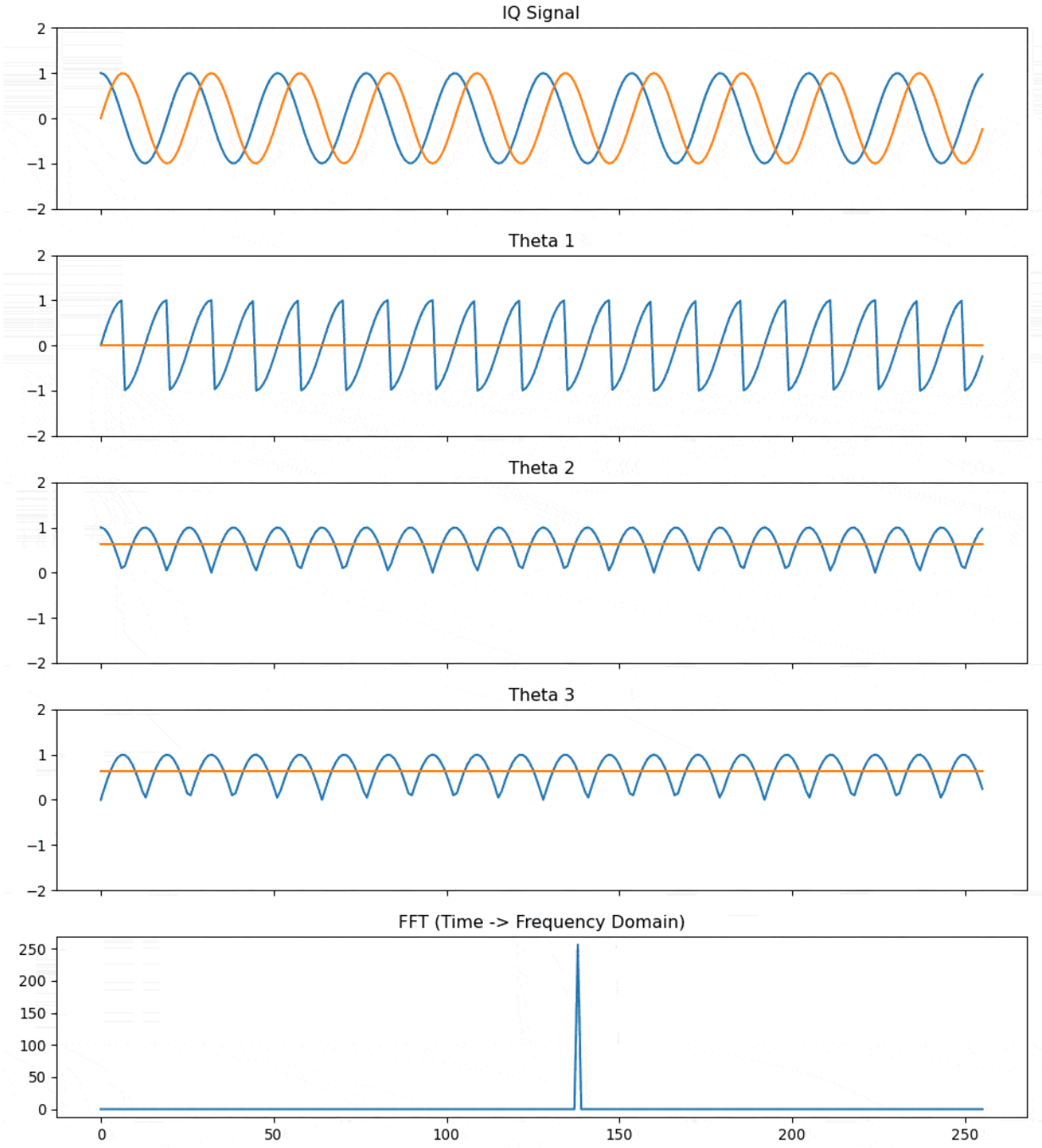


Once these quantities are estimated, we smooth the results using an exponential smoothing filter. We can calculate how an adjustment to the I and Q channels. The calculations are a little more complex than the theta values, and we need to calculate a square root. This isn't too much of an issue because we only need to calculate the adjustment once for each block of data. The whole process can be completed in a few lines of C++ code.

```
static uint16_t index = 0;
static int32_t theta1 = 0;
static int32_t theta2 = 0;
static int32_t theta3 = 0;

theta1 += ((i < 0) ? -q : q);
theta2 += ((i < 0) ? -i : i);
theta3 += ((q < 0) ? -q : q);

static int32_t c1 = 0;
static int32_t c2 = 0;
if (++index == 512)
{
  static int64_t theta1_filtered = 0;
  static int64_t theta2_filtered = 0;
```

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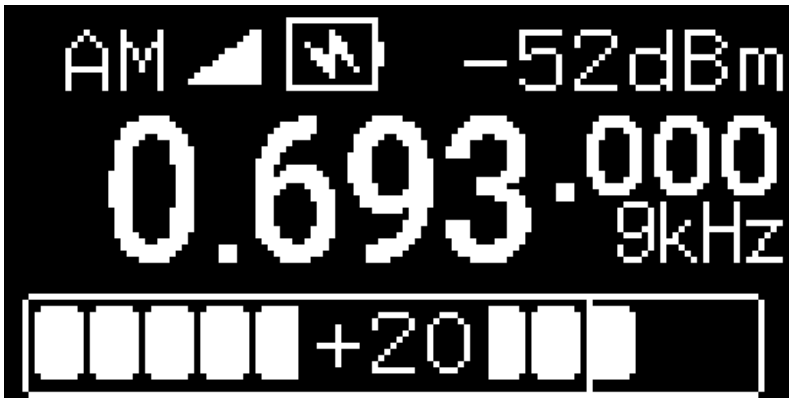
```
        static int64_t theta3_filtered = 0;
        theta1_filtered = theta1_filtered - (theta1_filtered >> 5) + (-theta1 >> 5);
        theta2_filtered = theta2_filtered - (theta2_filtered >> 5) + (theta2 >> 5);
        theta3_filtered = theta3_filtered - (theta3_filtered >> 5) + (theta3 >> 5);

        //try to constrain square to less than 32 bits.
        //Assue that i/q used full int16_t range.
        //Accumulating 512 samples adds 9 bits of growth, so remove 18 after square.
        const int64_t theta1_squared = (theta1_filtered * theta1_filtered) >> 18;
        const int64_t theta2_squared = (theta2_filtered * theta2_filtered) >> 18;
        const int64_t theta3_squared = (theta3_filtered * theta3_filtered) >> 18;

        c1 = (theta1_filtered << 15)/theta2_filtered;
        c2 = intsqrt(((theta3_squared - theta1_squared) << 30)/theta2_squared);

        theta1 = 0;
        theta2 = 0;
        theta3 = 0;
        index = 0;
    }

    q += ((int32_t)i * c1) >> 15;
    i = ((int32_t)i * c2) >> 15;
```

The solution works quite well, I tried to measure the performance improvements, but with the IQ correction enabled the signals have either disappeared below the noise floor, or are too small for the ADC to measure. This is a good result, in practice we won't be able we won't be able to hear any images with IQ correction enabled!

#####Insert before and after clip here

## UI Enhancements

Thanks to Mariusz Ryndzionek and Penfold42 for numerous improvements to the UI!




There have been numerous improvements to the user interface including nicer, easier to read fonts, nicer graphical elements and a cute splash screen!

One of my favourite improvements is the introduction of multiple "views" which can be accessed by pressing the back button on the home screen. This gives the option to control and status information in different ways without cluttering up the home screen. There is even a spectrum-scope with zoom and a dithered waterfall. I'm amazed at just how much can be achieved using a 128x64 display with only two colours.

The UI is now using the u8g2 library, which provides a rich set of graphical tools, and loads of fonts. The u8g2 also supports loads of different display types which should make it easier to adapt the software to new displays in future.

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## Scan/Search

Thanks to Penfold42 for contributing the scanning/search functionality!

There are 2 scanning modes, frequency scan and memory scan. In frequency scan mode, the receiver searches for signals in the current band frequency range (e.g. 20m SSB band). In memory scan mode, the receiver searches memory channels for active signals.

In both modes, the squelch setting is used to determine the threshold level, if the signal strength exceeds the squelch threshold, the search is halted. Searching can be continued by rotating the encoder.

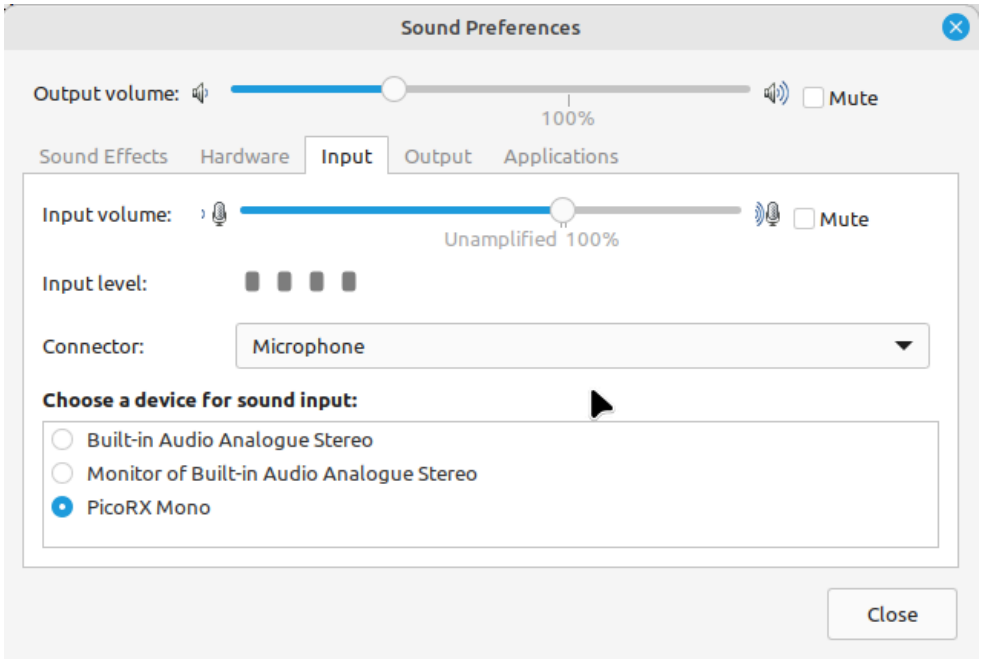
The encoder controls both the direction and speed of the search. The current signal strength and squelch level are indicated by a vertical bar on the right hand side.

## USB Audio

Thanks to Mariusz Ryndzionek for implementing USB audio!

The addition of USB audio makes it so much easier to make an audio connection to a host device. No messing around with audio cables, levels and sound cards. The audio is transferred digitally to the PC with no loss of information in the process.

The USB audio functionality is implemented using the tinyUSB library, which is a fairly complex library to use (but worth the effort). Although the library has loads of examples, the documentation is a bit thin. The USB audio code implements a standard UAC2 audio class, so the pi-pico appears as an audio microphone when connected to the host.




This allows a host device to easily make audio recordings (e.g. using Audacity), and is compatible wi software such as wsjtx, fldigi and QSSTV. When combined with USB cat control allows a fully functional

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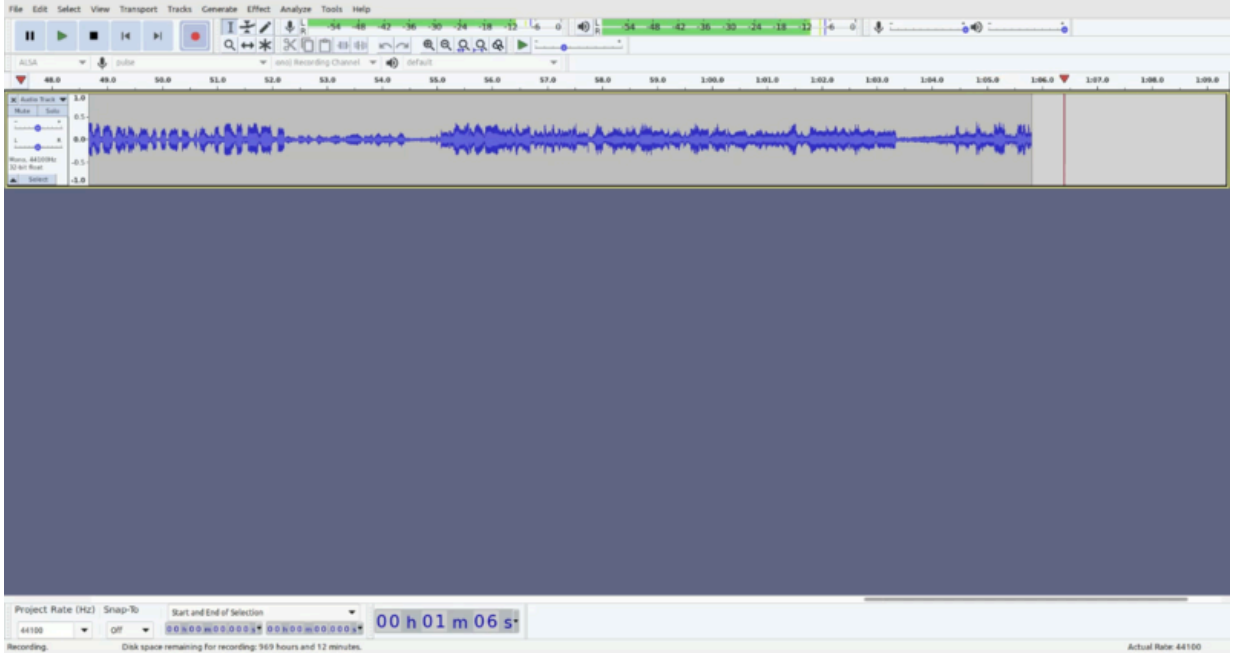
PC connection using only a single USB cable. The direct digital audio connection provides superior audio quality compared to an analogue connection using a sound card.




# USB CAT Control

CAT control is provided through a USB serial port interface. The Pi Pico Rx emulates a subset Kenwood TS-480.

The neat thing about USB is that a device can support multiple endpoints. This means that with a single USB connection, the receiver can appear as both a USB microphone and a virtual serial port. This makes it much easier to connect a radio to a PC or other host device. The CAT interface allows the receiver to be controlled via a host device by software such as grig, wsjtx and fldigi.

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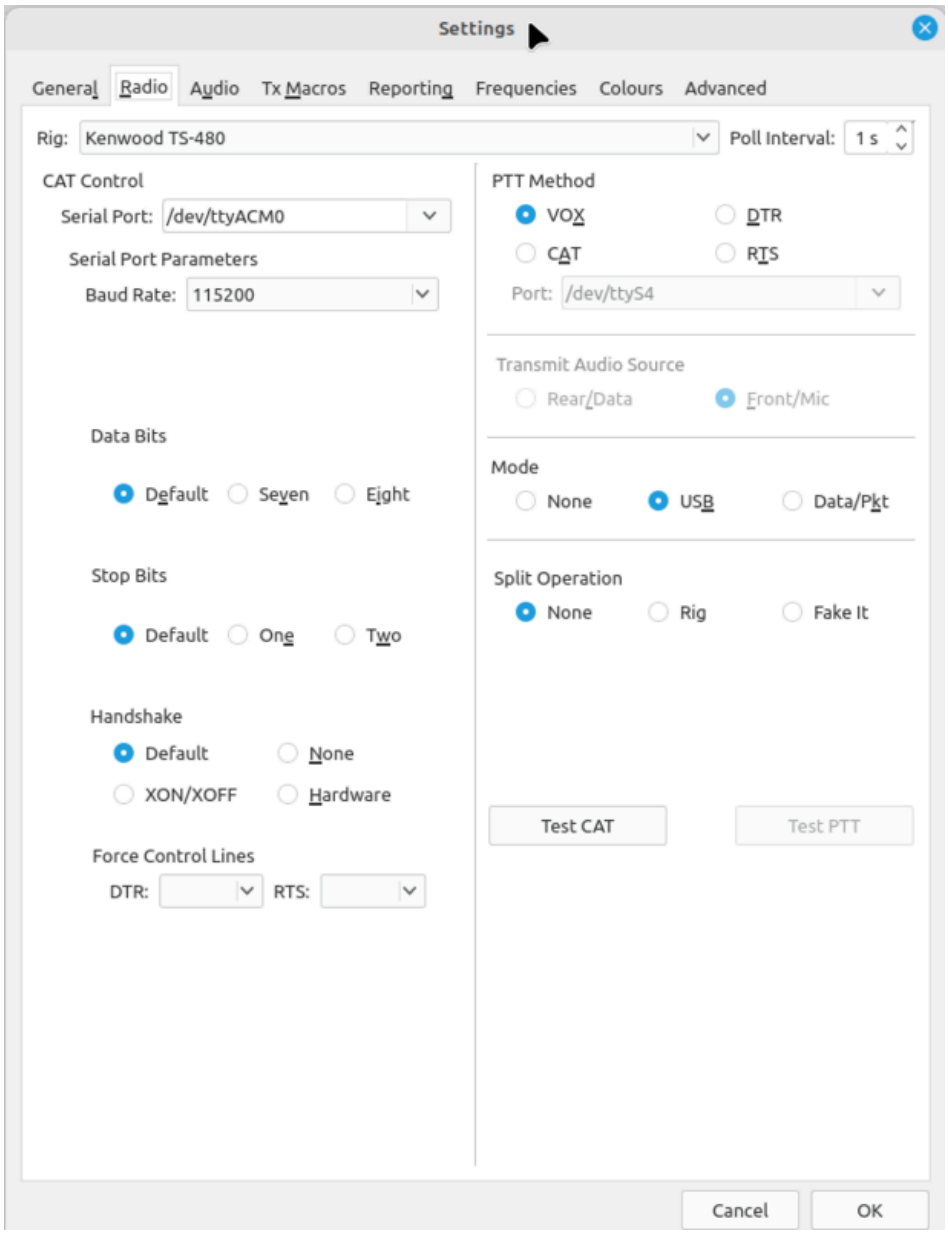


The combination of USB Audio and CAT control opens up a lot of possibilities for headless devices without a UI. It should be possible to build a simple adaptor that allows an antenna to be connected to a PC, laptop or smartphone for easy portable operation with minimal components. A smartphone has a larger screen and makes it possible to build a much richer user interface than a standalone device. A smartphone also has a lot more computing power, opening up many more possibilities for digital modes and internet connectivity.

# Calibration

Thanks to Penfold42 for contributing the signal strength calibration functionality!

The signal strength meter and s-meter have a power scale which makes use of the theoretical gain of the QSD and op-amp. In the breadboard version, the op-amp has a gain of 57dB. However, constructors may choose different filters or preamplifiers, there will always be variations between theoretical gains and real-world values. It makes sense to have some way to adjust or calibrate the gain display on the screen. The gain can be calibrated using a signal generator with a known signal strength. I don't have a calibrated reference, but the tinySA can be configured to output a signal with a known strength. I found that the reading on my tinySA was quite close to the value reported on the signal strength meter, but with a bit of adjustment, the reading was bang on. In theory, you could use a calibrated instrument to perform the same checks, but the tinySA gives a good indicative value and it is certainly good enough for most purpose

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A similar calibration can be performed for the frequency. I have added a menu item that allows the oscillator error to be specified in parts per million (PPM). Again, we could perform the calibration using a signal

generator with a known frequency. Another method is to tune in to a radio station with a known frequency. WSJTX includes a nice frequency calibration feature that we can use for this purpose. I chose to tune in to BBC Radio 4 on 198kHz, the carrier is controlled by a rubidium clock. The 500kW transmitter is based at Droitwich, only a few miles away, so I don't have any issues getting a signal. I found that with my receiver, a setting of 5ppm gave me the best accuracy. This is well within the +/-30ppm tolerance of the pi-pico's crystal oscillator. With calibration, the accuracy of the receiver is within 1ppm.

## TFT Waterfall/Spectrum Scope

The small OLED displays are cheap, readily available and very simple to wire up. They are pretty efficient and use hardly any power. Ideal for a minimalist radio like this one! However a waterfall and spectrum scope is a nice addition to any receiver, and it is tempting to see just how far we can push a simple design. Don't worry though, the OLED display isn't going anywhere any time soon. The TFT-based waterfall is just an optional extra that can be added with just a few extra wires and can be enabled through a menu configuration.

The waterfall uses an ILI9341-based TFT display. There are many such displays available, and they come in a range of sizes. I used a 2.4-inch display which seems to be one of the more popular sizes, they are easily available and inexpensive. The receiver is compatible with 4-wire SPI-type displays.

There are a few spare pins left on the pi-pico, some of which can be configured to work with the dedicated SPI hardware. It is handy to keep an SPI port spare, as it allows the pico to connect to other SPI devices too.

Although the display is a write-only device that doesn't need the MISO card, it makes sense to reserve it for use with other peripherals. The display has a hardware reset input, but it is fairly common practice to hard wire the reset to VCC to save the extra IO pin. I use the software reset functionality, and it seems to be working OK so far.

| Signal | GPIO |
|---|---|
| MISO | 12 |
| CS | 13 |
| SCK | 14 |
| MOSI | 15 |
| DC | 11 |

Displays come in different flavours. The ones I have seen may be configured to work a 5V or 3.3V supply, a no-fit resistor pad can be shorted to bypass an onboard regulator for 3.3V use. It is quite important to check that the display you are using is powered by the right voltage. (And also that the pico is capable of supplying enough current for the display). As far as I know, the display IO is always 3.3v.

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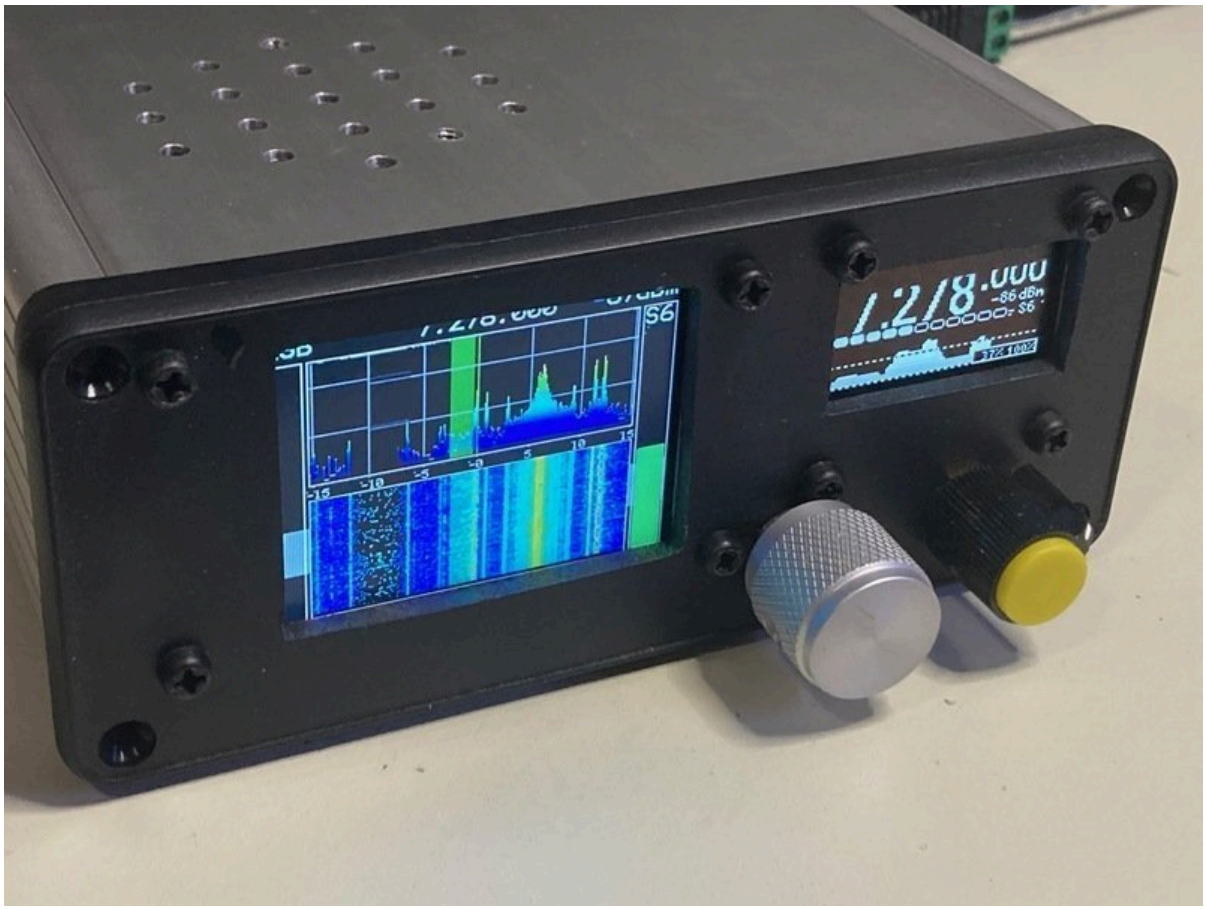
I used this library to drive the display. The nice thing about this display is that it is very simple and t code is easy to understand. It doesn't use any kind of frame buffer which is an advantage because the pico doesn't have enough spare RAM for this. It does mean that the SPI has to run at quite a high frequency to

achieve a reasonable refresh rate though. I have modified the driver quite heavily, to make drawing the waterfall and spectrum scope as efficient as possible. I also changed the text-drawing functions to allow them to use the same font as the OLED display.

## Hardware Configuration Options

Part of the fun of this project is that it isn't a finished product, more like a starting point for experimenters and constructors to build their own receiver. It makes sense to make the software as configurable as possible to support the widest range of receivers. There is now a hardware configuration that allows you to change many settings. Everything from the OLED display driver to the encoder direction and resolution. Wire the QSD backwards? - No problem, just switch I and Q in the hardware configuration menu. You can even change the frequency ranges of the band filters (if you have them), design a set of low-pass filters to build a general coverage receiver or add a set of bandpass filters for your favourite bands.

## Project Gallery




Pi Pico Rx Build with waterfall (courtesy Jim Reagan W0CHL)

A growing number of people have now completed a build of the breadboard receiver, and there is now a gallery of receiver builds. If you have built the pico breadboard receiver, I would love to hear from you! Gallery.

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